

Speech and Audio Systems

Project: Verified Study Buddy

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Visual Composition

Without opening the old lesson, explain one key idea from Visual Composition. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Prompt Evaluation

Without opening the old lesson, explain one key idea from Prompt Evaluation. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Examples and Rubrics

Without opening the old lesson, explain one key idea from Examples and Rubrics. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Trace the worked case for Speech pipelines and multimodal evidence. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Speech pipelines and multimodal evidence. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Changed independent case

Known audio says “nineteen minus six”. Transcription says “ninety minus six”; the calculator returns 84. Locate the first error and propose a check.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

Audio says 'sixteen divided by four'; transcript says 'sixty divided by four'; calculator returns 15. Which stage first fails? What would a useful confirmation show?

6. Solve the given case

A speech tool hears fifteen as fifty, then calculates correctly using fifty.

Where did the error enter?

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

A recording says fifteen, but the transcript says fifty. The summary repeats fifty. Where did the error enter, and what should be checked first?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 12 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Show transcripts and require confirmation before saving memory or using tools.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

Audio says 'sixteen divided by four'; transcript says 'sixty divided by four'; calculator returns 15. Which stage first fails? What would a useful confirmation show?

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Explain the mechanism

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

My reasoning and evidence

13. W2 Trace the worked case

Audio transcript reads fifteen boxes; a clear written inventory says fifty boxes. The original audio is unavailable and the assistant reports fifteen as certain. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

14. W3 Audit the limit

A report omits this qualification: A stage-level accuracy or authority policy does not establish every modality agrees or every downstream claim is supported. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

15. W4 Apply a new case independently

Transcript says room E; a current written approval says room F. The task requires current approved room. Written approval is declared authoritative; trace the response and the transcription limit.

My reasoning and evidence

16. W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: Speech and Audio Systems

A fictional learner dictates a calculation to a voice assistant.

Known audio phrase: “thirteen plus seven”. Transcript: “thirty plus seven”. Calculator returns 37.

Synthetic voice says “thirty-seven”. The example supplies ground truth; no real audio was analysed.

Speech and Audio Systems

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Recognise speech

Transcription changes thirteen into thirty.



2. Calculate text

$30+7=37$ is correct for the mistaken transcript.



3. Speak result

Speech synthesis repeats 37; fluent voice does not validate the input.



4. Confirm intended input

Read back the recognised expression; intended $13+7=20$.

CHANGE ONE THING

Correct the transcript to "thirteen plus seven" with all later stages unchanged.

Explain the changed result using the matching lesson.

Which stage was initially wrong, and which stage was correctly operating on a wrong input?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: Show Study Buddy recognised questions before calculating or retrieving important answers, especially numbers and dates.